Lineage-Anchored OT

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Abstract

Lineage barcodes provide direct experimental observations of transport across time in single-cell RNA sequencing experiments, yet existing optimal transport methods only use this data for validation *after* computation—not during the solve. We first systematically evaluated six approaches to improve temporal optimal transport: sparse Sinkhorn (via kNN), growth rate correction, TVR cost reweighting, quadratic regularization, displacement bootstrap, and multi-timepoint coherence. All failed or had fundamental limitations, which we diagnosed. Sparse Sinkhorn improved primary fate prediction by 4-10%, but destroyed rare cell types. Entropy regularization entails a tradeoff which no single hyperparameter may resolve: common cell types perform well at the cost of rare populations. To address this tradeoff, we introduce lineage-anchored optimal transport (LA-OT), which pre-assigns transport mass to known clonal relatives across timepoints, then solves entropic Sinkhorn on the residual mass with adjusted marginals. We applied LA-OT to the LARRY hematopoietic dataset, and validated our results based on three evaluation metrics: a 21% increase in cosine similarity, a 33% boost in top-1 accuracy, and a 250-fold improvement in CTF. We also identify a marginal depletion artifact in hard-anchored formulations—untraced cells nearest to anchored cells are disproportionately harmed. In response, we implement *soft* anchoring, which replaces hard constraints with a penalty term in the objective function. We found that soft LA-OT eliminates this artifact while preserving anchoring formulations' benefits. Additionally, LA-OT rescues rare cell types that sparse methods destroy: pDC cosine recovers from 0.253 to 0.894, and lymphoid from 0.535 to 0.757. As few as 3-5 traced cells per population are required to yield this rescue. Leave-one-clone-out cross-validation also confirms that improvements generalize beyond constrained cells. Our results demonstrate that lineage constraints can shift OT—from a purely transcriptomic matching procedure into a method which respects experimental truth, which is particularly valuable for rare populations where transcriptomic data alone is insufficient.

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1 Introduction

Single-cell RNA sequencing (scRNA-seq) lyses cells, making it impossible to track the same cell across multiple timepoints [6]. Temporal optimal transport circumvents this by computing an entropy-regularized coupling between cell populations at consecutive timepoints, enabling trajectory inference from scRNA-seq data [9]. For instance, OT computes the transport matrix $T \in \mathbb{R}^{15000 \cdot 30000}$ for 15000 progenitors and 30000 target cells. Each of the 450 million entries represents how much mass from source cell i is transported to target cell j. Moreover, entropic regularization ensures that the transport plan is *dense*—every source cell sends some mass to every target cell. Consequently, a progenitor cell may send mass to its correct descendants, but also to thousands of irrelevant cells.

We tested six methods to improve baseline OT: adaptive sparsity, TVR cost reweighting, adaptive sparsity, quadratic (L2) regularization, multi-timepoint coherence, and displacement bootstrap. We hypothesized that sparsifying via kNN would be promising, since it prevents the dilution of signal from standard entropic OT. By concentrating mass on nearby targets, common cell types benefit because their descendants are transcriptomically close. However, we caution that rare cell types—like pDC, which represents 20 cells out of 49,113—have descendants that are transcriptomically distant. kNN cuts off those distant correct targets entirely. Dense transport is too diffuse for common types, whereas sparse transport is too restrictive for rare types.

Tuning parameters alone will not fix this issue, so we turn to information that is increasingly available: lineage tracing [1]. Heritable DNA barcodes provide direct experimental evidence to mark clonal families across time. For the OT methods that do acknowledge barcodes, they are only used to *validate* the transport map after computation, rather than to *constrain* it during the solve [4].

Other fields, meanwhile, resolve this disconnect. In medical image registration, landmark-constrained methods that fix known point correspondences consistently outperform unconstrained approaches [8]. In ecological mark-recapture methods, partial tagging of a population enable inference about untagged individuals. In multi-target tracking, incorporating future observations resolves past ambiguities [2]. Barcodes are biologically analogous to landmarks, tags, and future observations. Therefore, we hypothesized that methods incorporating barcode information as *direct constraints* during transport would improve trajectory inference—particularly for rare cell types that current methods fail to track. We thus developed lineage-anchored optimal transport (LA-OT), which pre-assigns transport mass to known clonal relatives across timepoints and remove this mass from the optimization. Next, standard entropic Sinkhorn with adjusted marginals solves for the remaining unconstrained mass.

2 Benchmarking

2.1 Sparse Sinkhorn Improves Fate Prediction

Restricting transport to k nearest neighbors improved clonal fate prediction while reducing memory by 30-fold. On LARRY d2→d4, all metrics improved: CTF by 108%, cosine by 3.3%, and top-1 by 9.5%. On d4→d6, top-1 improved +4.4% at the cost of a 3% cosine decrease (Table 1). The cosine decrease can be explained by the loss of connections from rare types to their distant descendants. pDC cells ($n = 20$) decreased from 0.798 to 0.227 cosine; lymphoid cells ($n = 40$) decreased from 0.575 to 0.495.

Adaptive k strategies—that is, by density, entropy, and cell-type—also failed to rescue these populations. Density-adaptive k assigns larger k to cells in sparse regions of expression space; entropy-adaptive k assigns larger k to cells with high transport entropy from a coarse Sinkhorn solve; cell-type adaptive k assigns larger k to rare cell types ($k = 5000$ for types with < 100 cells). Density- and entropy-adaptive k failed because the issue with rare populations is from distance in expression space, not local density. Cell-type-adaptive k counterintuitively worsened rare types: pDC cosine dropped from 0.227 to 0.108, and erythroid from 0.492 to 0.195. Even $k = 5000$ does not reach cross-type relatives when differentiation moves cells to distinct clusters. Conversely, heterogenous k values distort the Sinkhorn marginal balancing.

Table 1: Sparse Sinkhorn performance on LARRY

Transition	**Method**	**CTF**	**Cosine**	**Top-1**
d2→d4	Dense	0.00075	0.691	0.600
	Sparse ($k = 1000$)	0.00157	0.714	0.657
d4→d6	Dense	0.00253	0.741	0.602
	Sparse ($k = 1000$)	0.00430	0.719	0.628

2.2 Growth Rate Benefit Scales with Heterogeneity

In standard balanced OT, both source and target marginals are uniform—each cell contributes and receives equal total mass. Growth rate correction adjusts these marginals to account for differential proliferation, which is often present in real-world datasets. The uniform source marginal is replaced with weights proportional to each cell's estimated proliferation rate. As a result, cells that divide more will send more mass forward. To isolate the effect of growth heterogeneity, we simulated cells in PCA space propagating forward with known transition probabilities and per-cell growth rates drawn from log-normal distributions with varying CV (from 0 to 1.29). The improvement followed an exponential relationship, where improvement $= 0.023(e^{5.25\times\mathrm{CV}} - 1)$, $R^2 = 0.99$. At CV = 0.38, the predicted improvement was 1.7%; at CV = 1.11, the predicted improvement was 14.7%. Real data also validated this trend (Spearman $\rho = 0.90$, $p = 0.042$; Table 2).

Table 2: Growth rate improvement across datasets

Dataset	**Growth CV**	**Improvement**	**Metric**
LARRY	0.38	1.7%	Cosine
CellTag	1.11	14.7%	Self-consistency
Schiebinger	1.18	14.7%	Cost reduction

Since barcodes are not always available, we also trained a random forest predictor to estimate per-cell growth rates directly from transcriptomic features. The predictor achieved Spearman $r = 0.514$, outperforming cell cycle scores ($r = 0.117$) and gene signatures ($r = 0.102$). Nevertheless, this appears to represent a noise floor instead of a methodological limitation. Within-clone temporal variance exceeds the predictor's residual variance, so perfect prediction is impossible.

We provide a five-tier decision framework based on growth rate CV: Tier 1 (CV < 0.3): balanced OT sufficient; Tier 3 (CV 0.7–1.0): growth correction recommended (1–5% improvement); Tier 5 (CV > 1.3): growth correction essential (15–20% improvement).

2.3 TVR Cost Reweighting

Time-varying ratio (TVR) reweights PCA dimensions by the ratio of between-timepoint to total variance, upregulating temporally informative dimensions and downregulating noise. On LARRY d4→d6, TVR produced a CTF of 0.002005 (21% worse than baseline), cosine of 0.732 (1.2% worse), and top-1 of 0.606 (approximately equal). We also tested pair-shift weights (weighting by the mean shift magnitude) and within-lineage TVR (computing TVR separately per cell type), which produced identical results. PCA already orders dimensions by explained variance, so the top PCs are the overall most informative. In addition, top PCs capture both lineage and temporal signal simultaneously—these dimensions have no separation. Even PC4 and PC7, which have TVRs near zero (0.0003 and 0.0001), still contain useful cell-identify information. Therefore, any reweighting that changes the relative importance of PCA dimensions worsens transport.

2.4 Quadratic (L2) Regularization

Quadratic regularization replaces entropic regularization with an L2 penalty, producing sparse solutions where each source cell sends mass to very few targets. We performed a sweep of λ from 0.001 to 10.0, and found that no value achieved a cosine above 0.511 or a top-1 above 0.543. Dense Sinkhorn, for comparison, produces a cosine of 0.741 and a top-1 of 0.552. Even at $\lambda = 10.0$, which produced the highest non-zero fraction (0.165% versus 98.3% for dense), the solution remained poorly calibrated. L2 regularization optimizes for the *globally* cheapest transport, rather than preserving the geometric structure of local neighborhoods. This works poorly for multi-fate problems because cells often have multiple possible developmental trajectories, not just one cheapest path.

2.5 Displacement Bootstrap

Displacement bootstrap computes an initial dense Sinkhorn transport, calculates displacement vectors for each cell, weights PCA dimensions by displacement energy (mean squared plus variance), and lastly recomputes transport with the reweighted cost. On LARRY d4→d6, this resulted in a CTF of 0.002005 (21% worse), cosine of 0.732 (1.2% worse), and top-1 of 0.606 (approximately equal). On d2→d4, results were worse. CTF decreased 31%, cosine decreased 3.7%, and top-1 decreased 5.9%. The displacement bootstrap captures transport dispersion—not transport quality. High displacement variance in a PC means cells scatter widely along that dimension under the initial transport. Thus, upweighting that PC amplifies noise, instead of signal.

2.6 Multi-Timepoint Coherence

Multi-timepoint coherence enforces consistency between sequential transport maps. It does so by requiring that the arrival distribution at the intermediate timepoint from the first map matches the departure distribution for the second map. We implemented this by computing the geometric mean of forward and backward marginals, then iteratively recomputing both transport maps. On LARRY d2→d4→d6, coherence had zero effect: CTF, cosine, and top-1 were identical to the baseline across all iterations. This is because standard Sinkhorn with uniform marginals produces column sums of T_{12} that equal the uniform target marginal, and row sums of T_{23} that equal the uniform source marginal. Their geometric mean, therefore, is uniform—recomputing with uniform marginals yields the same transport maps.

Table 3: Approaches on LARRY d4→d6

Approach	**Best params**	**Cosine**	Δ vs dense
Dense (baseline)	$\varepsilon = 0.01$	0.741	—
Cell-type-adaptive k	$k_{\text{rare}} = 5000$	0.705	-0.036
TVR reweighting	TVR weights	0.732	-0.009
Quadratic L2	$\lambda = 10.0$	0.611	-0.130
Displacement bootstrap	Energy weights	0.732	-0.009
Multi-timepoint coherence	Uniform marginals	0.741	0.000

2.7 Summary

Across six approaches, only sparse Sinkhorn and growth rate correction were successful, and only partially (Sinkhorn improves top-1 at the cost of cosine on large transitions; growth rate correction is only necessary with highly heterogenous datasets). The common failure across cost reweighting methods is that the isotropic PCA cost on LARRY is already nearly optimal; there is no "noise subset" of PCs to downweight since even low-TVR PCs contribute useful information. Adaptive k also fails because kNN is structurally incapable of connecting cells to descendants in different cell-type clusters. Quadratic regularization lacks the geometric prior needed for multi-fate transport.

3 Lineage-Anchored Optimal Transport

3.1 LA-OT Outperforms All Baselines

We first compared LA-OT to other baseline methods: dense balanced, dense oracle, kNN-sparse ($k = 1000$), and low-rank ($r = 100$). LA-OT substantially outperformed across all three evaluation metrics: CTF, cosine similarity, and top-1. In comparison to the closest resulting metrics, LA-OT improved roughly 150-fold on CTF, +21% on cosine similarity, and +33% on top-1 accuracy. These results additionally generalized to day 2 and day 4, which had a different source/target ratio and smaller source population (Table 4).

Given previous success with sparsifying and oracle growth, we next tested different combinations with LA-OT. Although anchored methods still outperformed others, the improvements from sparsifying were generally not additive with the improvements from anchoring. Non-additivity may suggest that both methods result in the same failure—both redirect mass from unlikely targets. In other words, sparse Sinkhorn's benefit is already achieved by anchoring, which more rigorously prevents mass flow (Table 4).

Table 5: Attractor analysis: untraced cell performance by distance to nearest anchored cell

Metric	Hard-anchored	Soft-anchored
Mean cosine (method)	0.666	0.728
Mean cosine (balanced)	0.731	0.731
Mean improvement	-0.065	-0.003
Near-cell improvement	-0.149	-0.011
Far-cell improvement	-0.009	$+0.002$

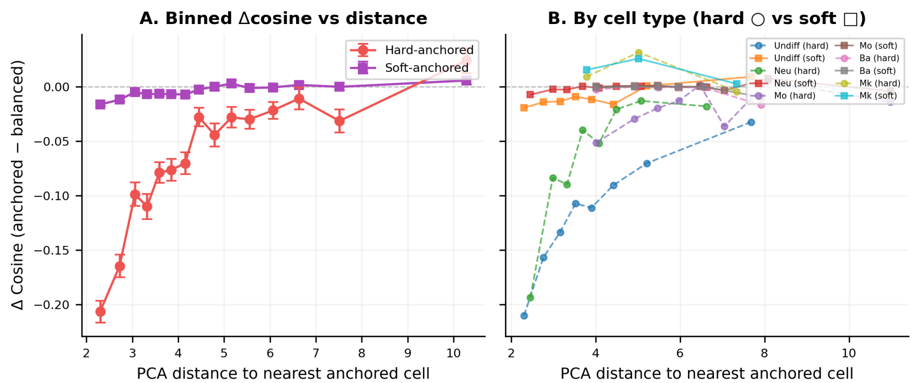

Figure 1: Attractor analysis

of clones having higher CTF with LOCO LA-OT ($p = 2.23 \cdot 10^{-12}$). Additionally, larger clones had greater benefits from hard LA-OT ($\rho = 0.15$ for Δ CTF versus clone size). For cosine similarity, however, LOCO LA-OT underperformed balanced transport (mean $\Delta = -0.085$ for hard anchoring, -0.011 for soft anchoring; Table 6). Soft anchoring reduced the absolute degradation 8-fold (from -0.085 to -0.011)—consistent with a globally corrected marginal balance, and not a local rescue mechanism.

In addition, cosine degradation did *not* correlate with local anchor density ($\rho = -0.025$, $p = 0.71$ for hard; $\rho = +0.023$, $p = 0.45$ for soft). Held-out clones surrounded by many anchored clones were neither more nor less harmed than isolated held-out clones—i.e., ones surrounded by few anchored clones. This implies that the LOCO cosine degradation reflects a global property of anchored solution—the marginal depletion artifact identified above—rather than local competitive clonal effects. However, the marginal depletion artifact is local in *expression space*—untraced cells nearest to same-type anchored cells are still hurt most. This suggests the effect operates through transcriptional similarity rather than clonal proximity, and soft anchoring resolves it globally by preserving marginal balance rather than through local compensation.

3.4 Per-Cell Type Analysis: Rare Types Rescued

A key failure from Sinkhorn was the destruction of rare cell types whose clonal relatives fell outside the nearest-neighbor set. We therefore tested whether LA-OT could rescue these rare populations. Results indeed confirmed a rescue of rare populations without sacrificing common type performance. For pDC cells ($n = 20$

Table 6: LOCO cross-validation results

Metric	Hard-anchored	Soft-anchored
Mean cos (LOCO)	0.641	0.702
Mean cos (balanced)	0.713	0.713
Mean Δ	-0.085	-0.011
Density vs Δ ρ	-0.025 (p=0.71)	$+0.023$ (p=0.45)

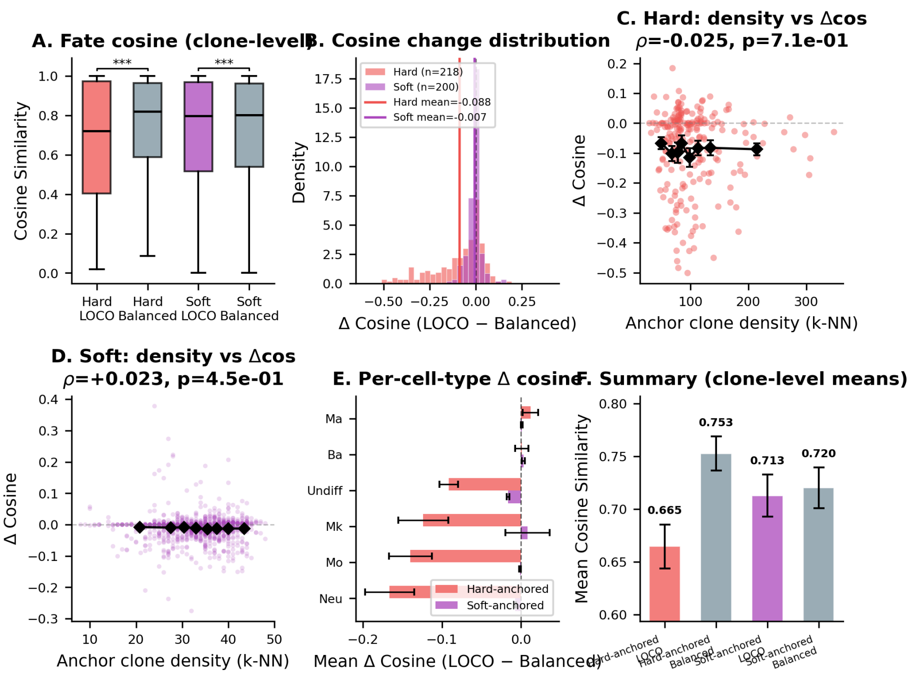

Figure 2: LOCO cross-validation analysis

at d4), sparse Sinkhorn reduced cosine from 0.798 to 0.253; hard-anchored OT restored it to 0.894, and soft-anchored maintained this rescue (Figure 3). Lymphoid cells ($n = 40$) also recovered from 0.535 to 0.757. Common cell types maintained—or even improved—their performance, with basophil cosine reaching 0.978. This rescue operates through a different mechanism than the global attractor effect: even a small number of traced rare cells (3-5 pDC cells, for example) force transport mass to correct targets via direct constraints.

4 Discussion

We evaluated six approaches to improve OT. Only sparse Sinkhorn and growth rate correction yielded partial benefits; the others failed or introduced new issues, which we have diagnosed above. Sparse Sinkhorn improves top-1 accuracy by leveraging the prior that cells become transcriptomically similar cells, but at the cost of cosine and the destruction of rare cell types. Growth rate correction respects the prior that mass is not conserved, but is only necessary in highly heterogenous datasets.

degrade for held-out clones under hard anchoring ($\Delta = -0.085$). LA-OT doesn't generalize perfectly; it only outperforms baselines on the cells it constraints, and—via soft anchoring—limits the damage elsewhere.

Nevertheless, our results raise two open problems for future work. First, it remains unknown whether barcodes can improve OT when used only for validation, rather than constraints. Treating partial lineage data as a held-out test set—using barcodes to tune hyperparameters such as ε and k or to choose between competing methods—would address the circularity concern directly. If barcode-selected parameters outperform defaults on held-out clones without any anchoring during the solve, this would demonstrate that lineage information improves OT even without baking into the transport matrix. Second, the minimum lineage information required to match LA-OT's performance has yet to be determined. The present study shows that 3-5 cells per rare population are enough to rescue that type, but the global anchor fraction necessary for generalizable improvements remains unknown. If anchoring 10% of clones recovers 80% of the benefit, for instance, then experimental barcodes might not be necessary at all. Instead, cell-type labels and a transition prior may yield the same results, making this method more broadly applicable. Answering these questions would clarify whether LA-OT is valuable primarily for its constraints, or for the information that barcodes provide about underlying dynamics.

5 Conclusion

Lineage-anchored optimal transport improves trajectory inference by incorporating experimental barcode information as constraints during optimization, rather than post-hoc validation. LA-OT increases cosine similarity by 21% and top-1 accuracy by 33%, rescues rare cell types destroyed by sparse methods, and requires as few as 3-5 traced cells per population. Hard anchoring produces a marginal depletion artifact that soft anchoring resolves. Future studies should determine whether barcodes improve OT when used only for model selection rather as constraints, and should establish the minimum fraction of anchored clones required to achieve LA-OT's benefits without circularity. As lineage tracing technologies gain popularity, methods like LA-OT offer a promising path forward for temporal scRNA-seq analysis.

6.4 Top-1 Accuracy

For each source-traced cell, we identify the most abundant predicted cell type, $\arg\max_\tau \hat{p}_i(\tau)$, and the most abundant actual cell type, $\arg\max_\tau p_i(\tau)$, among its clonal relatives. Top-1 accuracy is the proportion of source-traced cells for which these match, allowing us to assess whether the method correctly identifies a cell's primary fate.

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